Decoding transposable element-derived cis-regulatory and RNA functional elements in long non-coding RNAs

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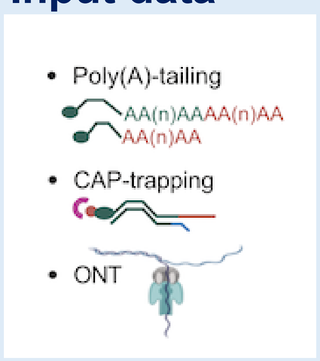
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Aim

Characterise how transposable elements (TEs) build cis-regulatory and RNA functional elements in human transcripts.

Why CFC-seq? TEs make up almost half of the genome, but short reads cannot resolve repetitive TE families. Cap-trapping + poly(A)-tailing long reads give unambiguous, full-length transcript models with precise TSSs.

Input data

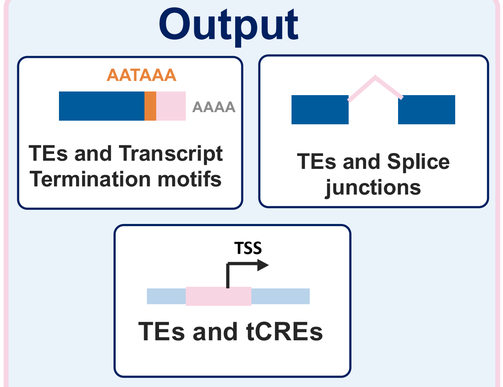


- CFC-seq full-length transcript models
  Libraries : HEK cells,THP1_Series, Neuron_Series
- Genomic TE annotation (RepeatMasker, hg38)
- tCREs defined from cell-type-specific TSSs

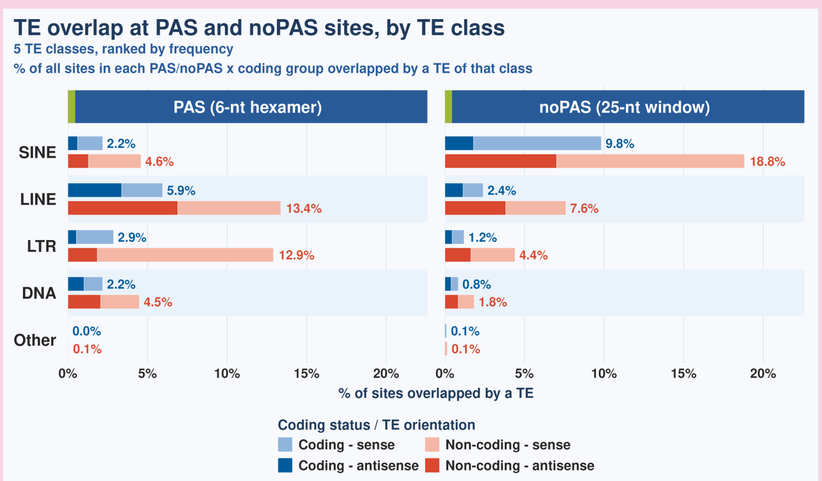
Methods

Step	Approach
tCRE calling	Cell-type-specific TSSs → promoter-like, enhancer-like, CTCF-like and other
Transcript models	Coding and non-coding, by CPAT
TE intersection	Splicing Junctions and PAS motif
Enrichment	Fisher's exact test, FDR

Output



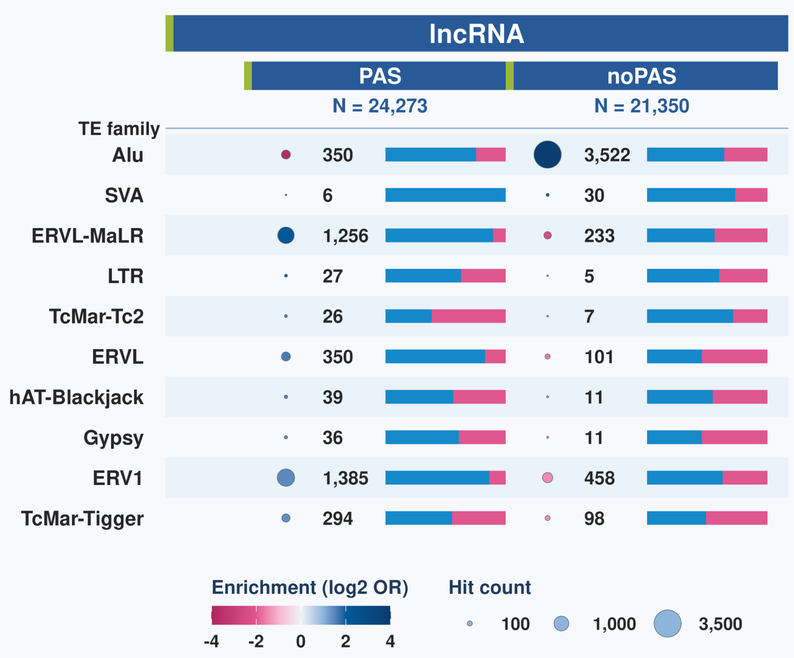
LINE and LTR elements are more common among PAS hexamers, while SINEs dominate noPAS sites; in lncRNAs, ERVL-MaLR and ERV1 show more PAS hits, whereas Alu shows more noPAS hits.



Blue = coding | **Red** = non-coding | **Light shade** = sense | **Dark shade** = antisense
Number at bar end = total % (sense + antisense)
Both panels share one axis, so bar lengths are directly comparable

TE families discriminate PAS from noPAS lncRNA sites

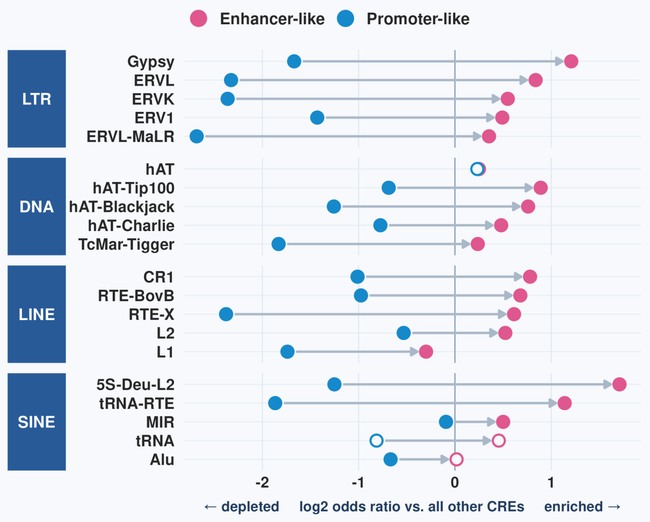
Top 10 significant families ranked by effect size (|log2 OR|), adjusted p < 0.05



Blue = enriched in the column group | **Pink** = depleted | **Light grey** = about equal
Bubble area & number = hit count | **All families:** adjusted p < 0.05
Bar beside each count: **sense** vs **antisense** share of TE overlaps

Transposable-element enrichment in enhancer-like vs promoter-like CREs

16 of 20 TE families are enriched in enhancer-like CREs, while 18 of 20 are depleted in promoter-like CREs



How to read

- **Pink** = enhancer-like
- **Blue** = promoter-like
- **Filled dot**: FDR < 0.05
- **Hollow dot**: not significant
- **Arrow**: promoter to enhancer

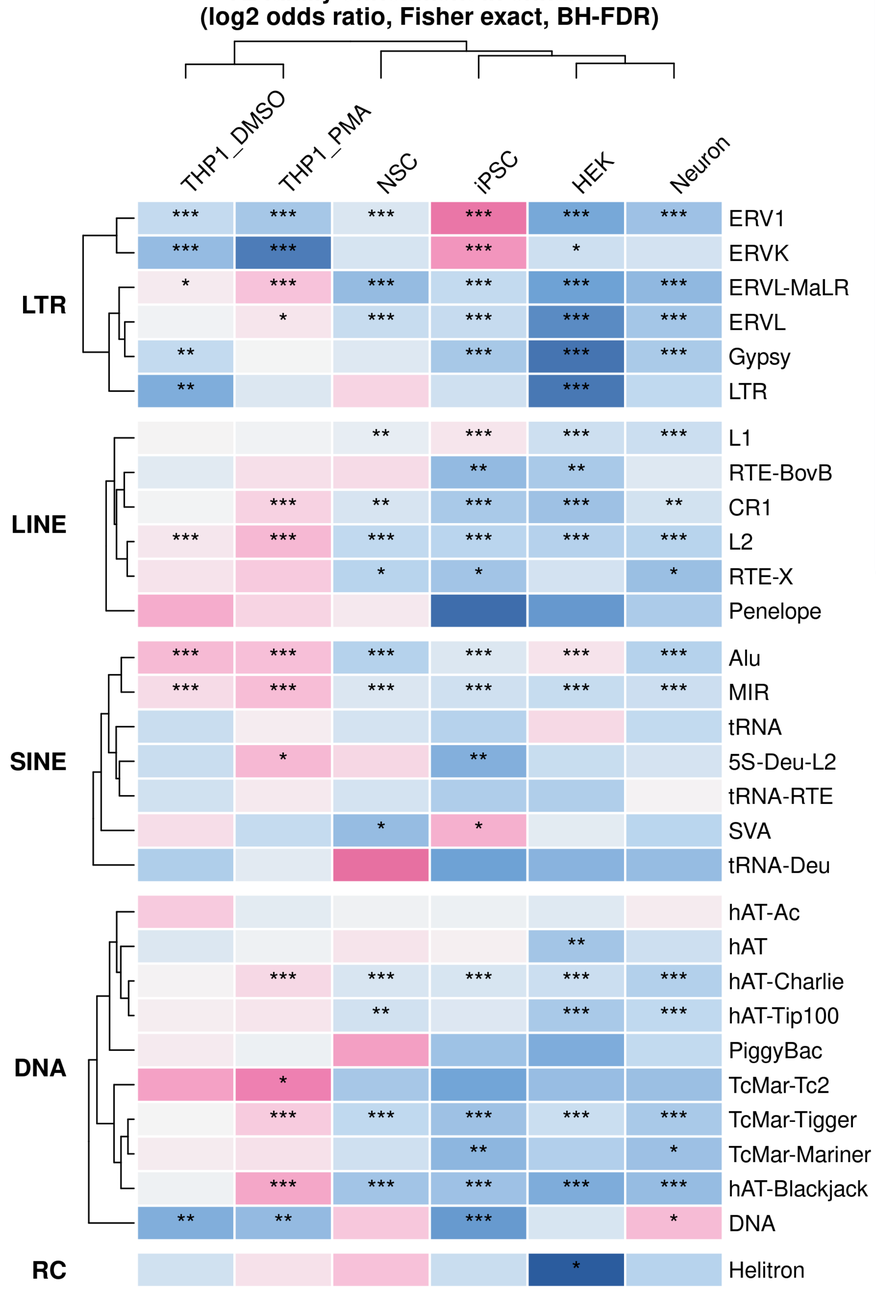
Data and test

Enhancer-like n = 56,599
Promoter-like n = 30,544 CREs

Top 5 most abundant families per TE class

Fisher's exact test (CRE-level), BH-corrected

TE family enrichment in enhancers (log2 odds ratio, Fisher exact, BH-FDR)

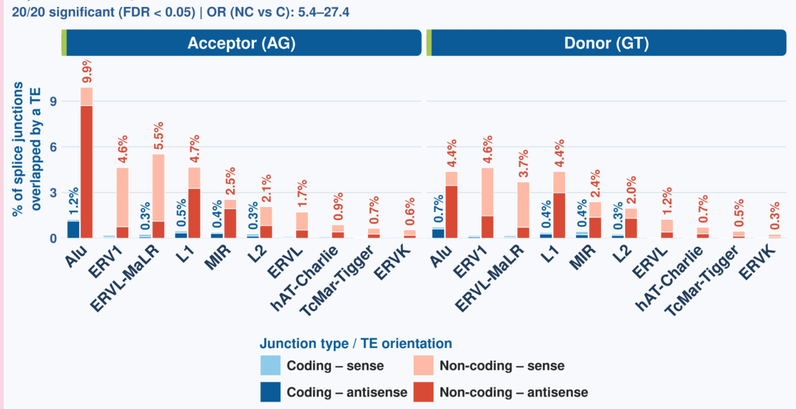


Rows are split by class. Within each class, families are clustered by enrichment profile (Ward.D2 on $\log_2$ OR). Blue indicates depletion ($\log_2$ OR < 0), red indicates enrichment ($\log_2$ OR > 0), and gray indicates not tested (families overlapping with fewer than 10 enhancers).
* FDR < 0.05; ** FDR < 0.01; *** FDR < 0.001.

TE-overlapped splice junctions are more common in non-coding genes (OR 5.4–27.4), with Alu reaching ~9.9% of acceptor sites; in lncRNAs, LTR sites are mostly sense-oriented, while SINE and LINE sites are mostly antisense.

TEs at splice junctions: coding (C) vs non-coding (NC) genes

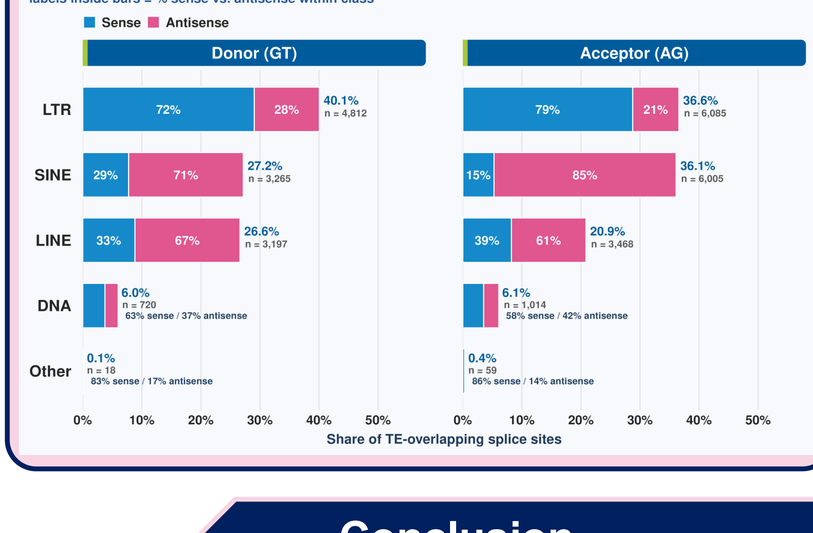
Top 10 TE families | C n = 173,726, NC n = 48,040
20/20 significant (FDR < 0.05) | OR (NC vs C): 5.4–27.4



Blue = coding (C) | **Red** = non-coding (NC) | **Light shade** = sense | **Dark shade** = antisense
Number above bar = total % (sense + antisense)
% of all C / all NC splice junctions overlapped by a TE of that family

TE classes at splice donor and acceptor sites in lncRNA

Bar length = class share of all TE-overlapping sites
labels inside bars = % sense vs. antisense within class



Conclusion

- **Full-length, long-read transcript models from CFC-Seq allow unambiguous mapping of TE-derived sequences.**
- **Over 20% of tCREs overlap TEs. They are enriched in enhancer-like tCREs and depleted in promoter-like tCREs, in a cell-type-specific manner.**
- **TEs shape transcript architecture. They contribute to splice junctions and PAS motifs. TE-overlapped splice junctions are more common in non-coding than coding genes, and with family-specific sense/antisense orientation in lncRNAs.**

Our results suggest that TEs are functional RNA building blocks. Rather than passive evolutionary remnants, can TEs act as modular elements of the human transcriptome's regulatory and RNA architecture?



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References:

1) Yip CW, Parr C, Takahashi H, et al. Genomic codes governing enhancer RNA fate. bioRxiv [preprint]. 2026 Jun 20. doi:10.1101/2024.10.31.620483.
2) Moody J, Kouno T, Chang JC, Ando Y, Carninci P, Shin JW, et al. SCAFE: a software suite for analysis of transcribed cis-regulatory elements in single cells. *Bioinformatics*. 2022;38(22):5126–5128. doi:10.1093/bioinformatics/btac644.
3) ENCODE Project Consortium. Expanded encyclopaedias of DNA elements in the human and mouse genomes. *Nature*. 2020;583:699–710. doi:10.1038/s41586-020-2493-4.